

PARTNERSHIP PROGRAM COMPUTER SCIENCE & ENGINEERING

Partnership Program | Adaptive Sports Center | Jeni Blacklock | Greg Vanderbeek

MCEN4026: Manufacturing Processes and Systems 2022 - 2023

Background

The Mechanical Engineering students in the Rady Partnership Program have been challenged to create an adaptive gravel bike for the Adaptive Sports Center (Adaptive SC) located in Crested Butte, CO. Adaptive SC helps enhance the quality of life of people with disabilities through outdoor adventurous activities. This bike will be donated to one of Adaptive SC's many athletes.

Key Requirements

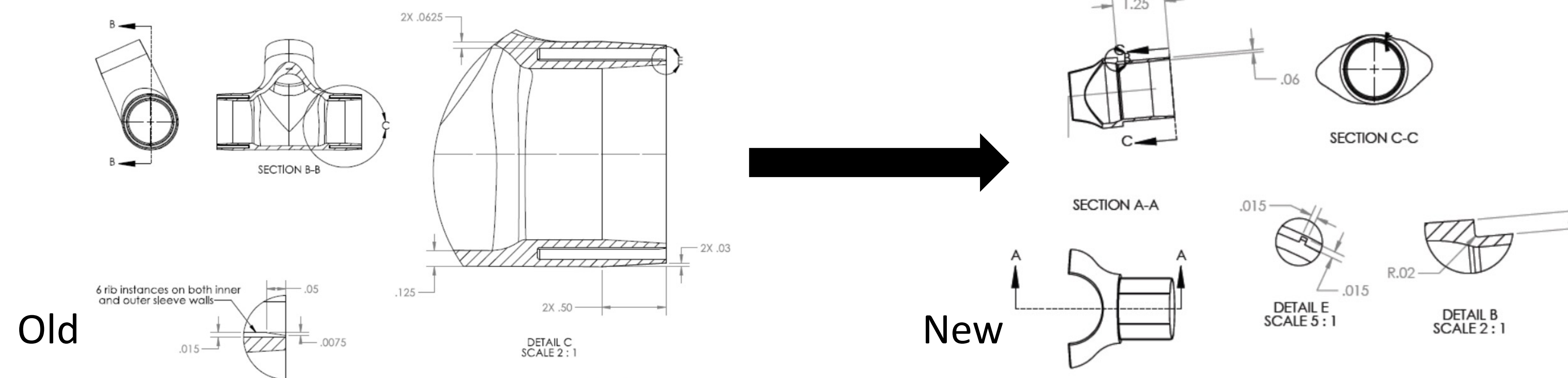
- Suitable for an adaptive athlete
- Weight Requirement: Max 250 lbs.
- Height Requirements: Max 6 ft., Min 5 ft.
- Ground Clearance: Min 7 in.
- Front wheel drive



Lug Design

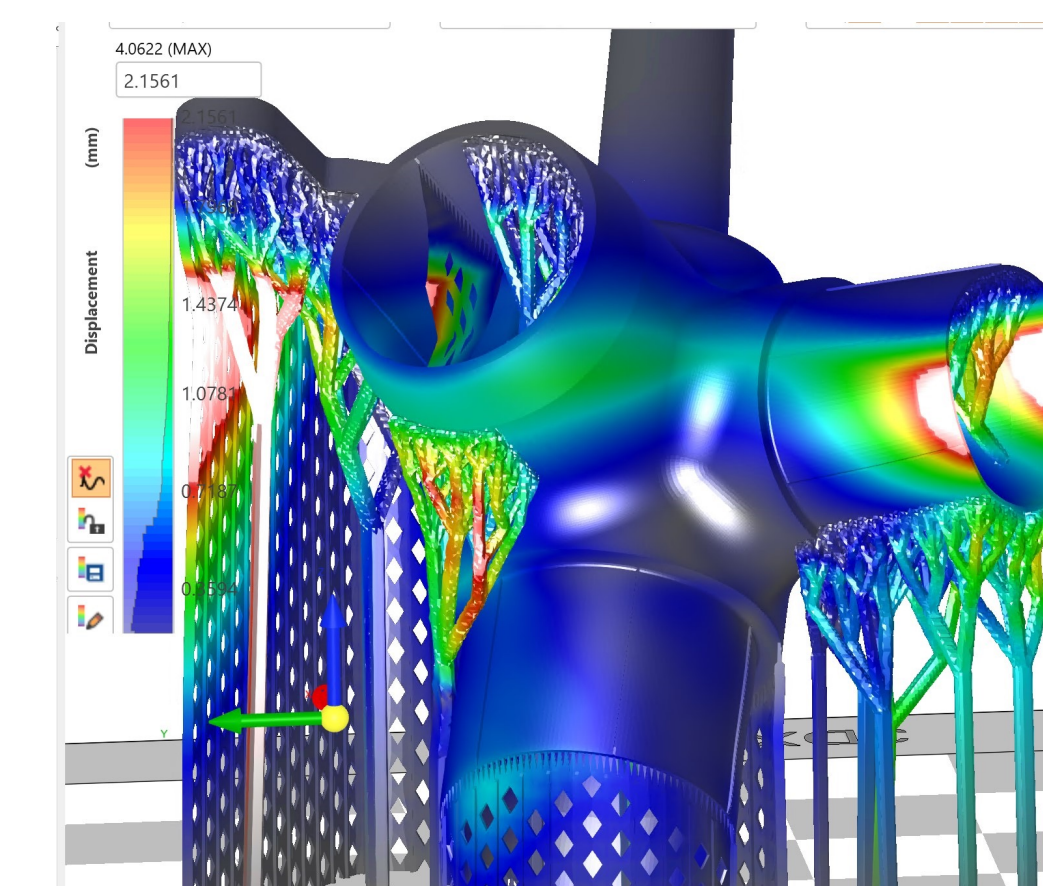
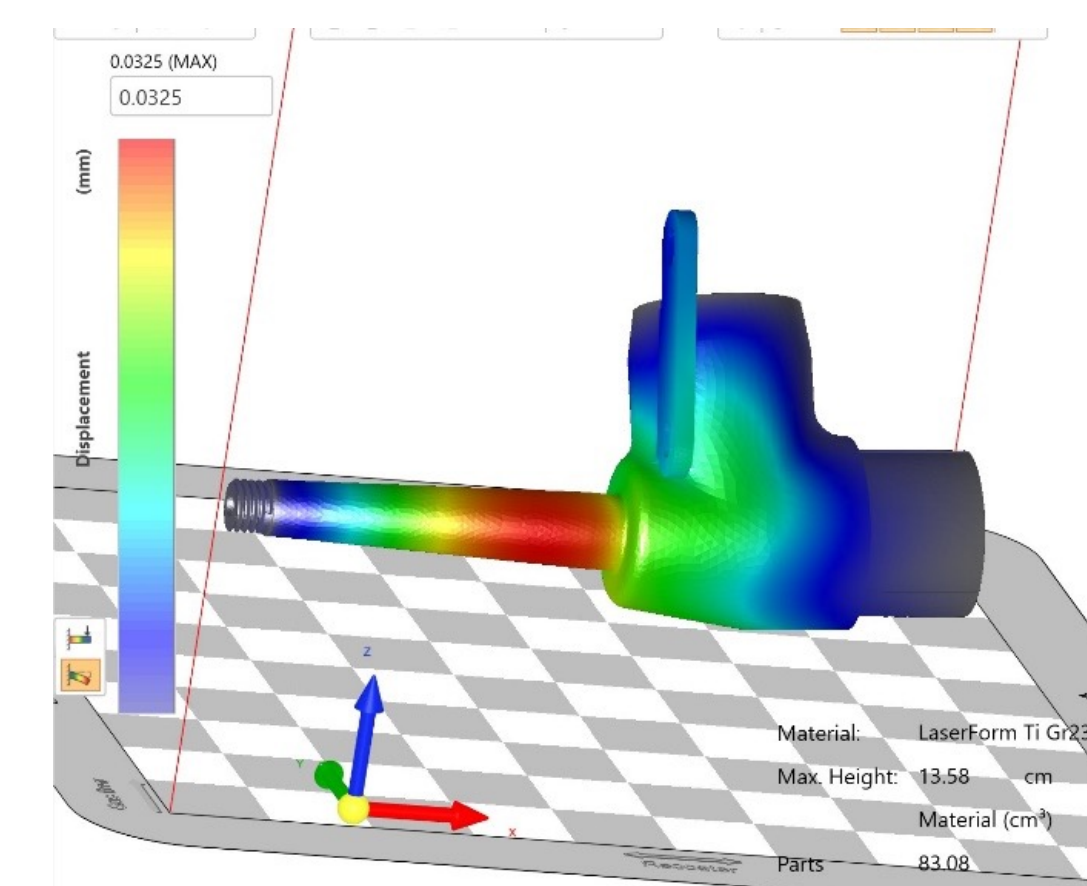
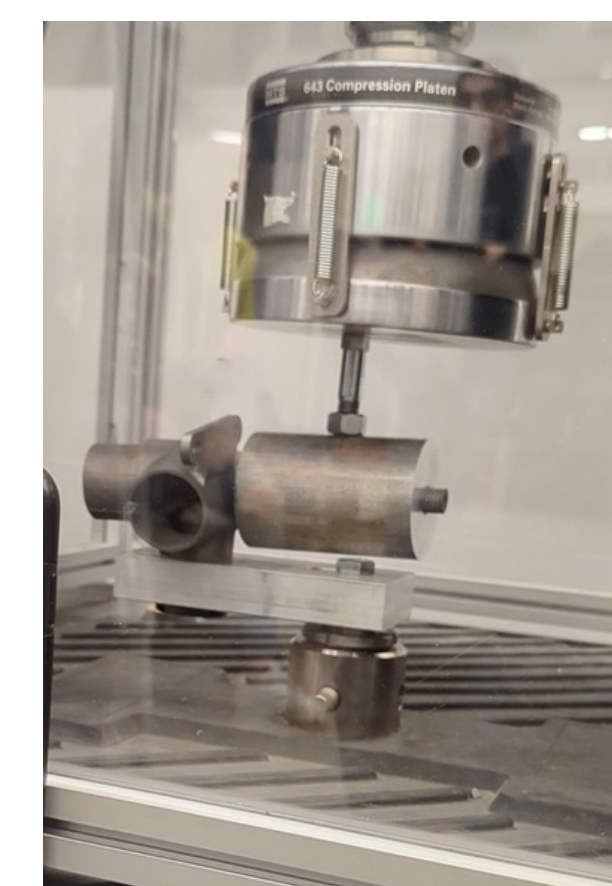


- Links carbon tubes together
- Bonded to carbon with high strength epoxy
- Bonding ridges were implemented to allow for even spacing of epoxy
- Went from double lap lug tube joint to a single lap to ease assembly process
- Printed then chased threads for connections



Testing

- To confirm FEA strength analysis rear axel lug was tested in the materials testing system (MTS)
- Max load reached was 3600lbs giving this lug a safety factor of over 10
- Thermal analysis of 3D printed titanium lugs



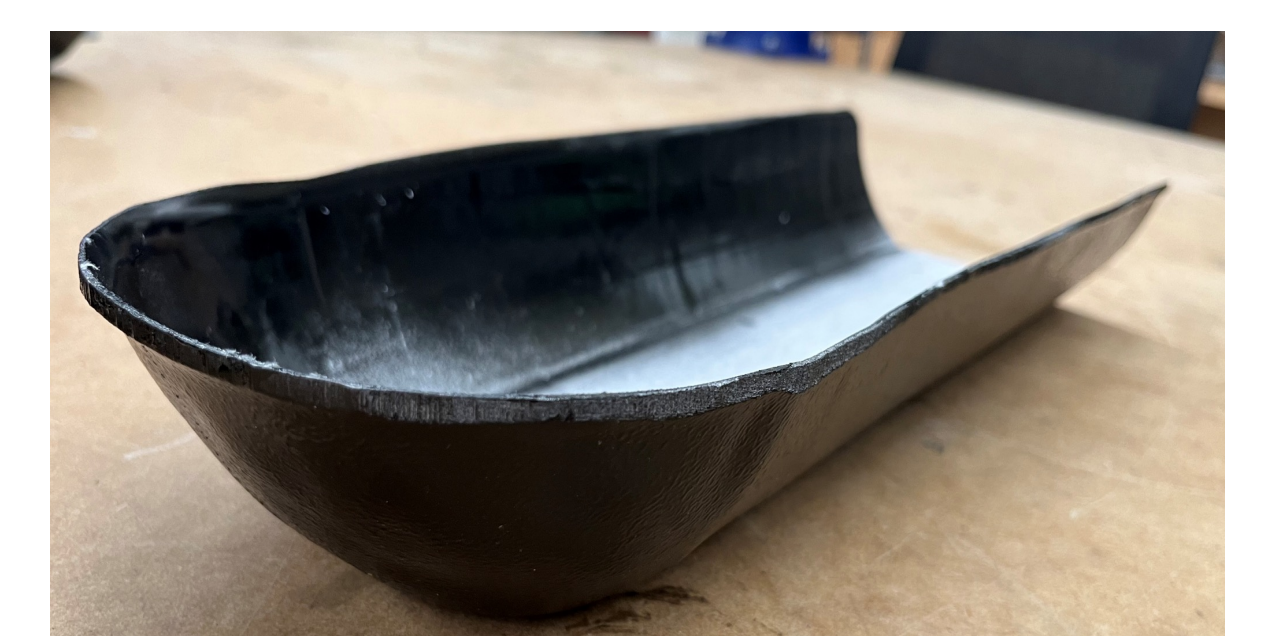
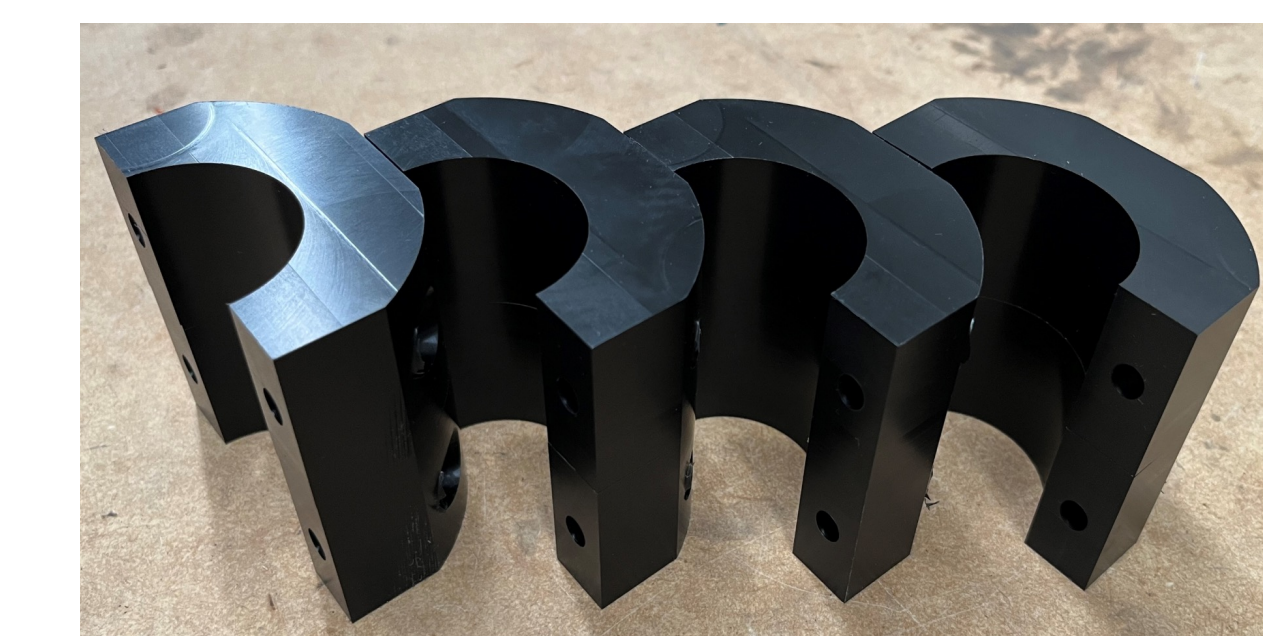
Braking

- Hydraulic brakes
- Dual rear and front brake calipers
- Locking feature on rear brakes
- Responsible for the handlebars
 - Bored out, welded, and powder coated aluminum to connect with a mountain bike pedal spindle



Ergonomics

- Padded knee baskets with channels for debris to settle
- Knee baskets thermoformed from Kydex
 - Tested for tear out
 - Machined Delrin mounting brackets to interface with carbon fiber frame
- Removable foam inserts
- No latex, repeated surgeries leads to latex allergy
- Adjustable seat design from Crazy Creek and NRS straps
- Rock guard to prevent debris from getting into the knee baskets



Drivetrain

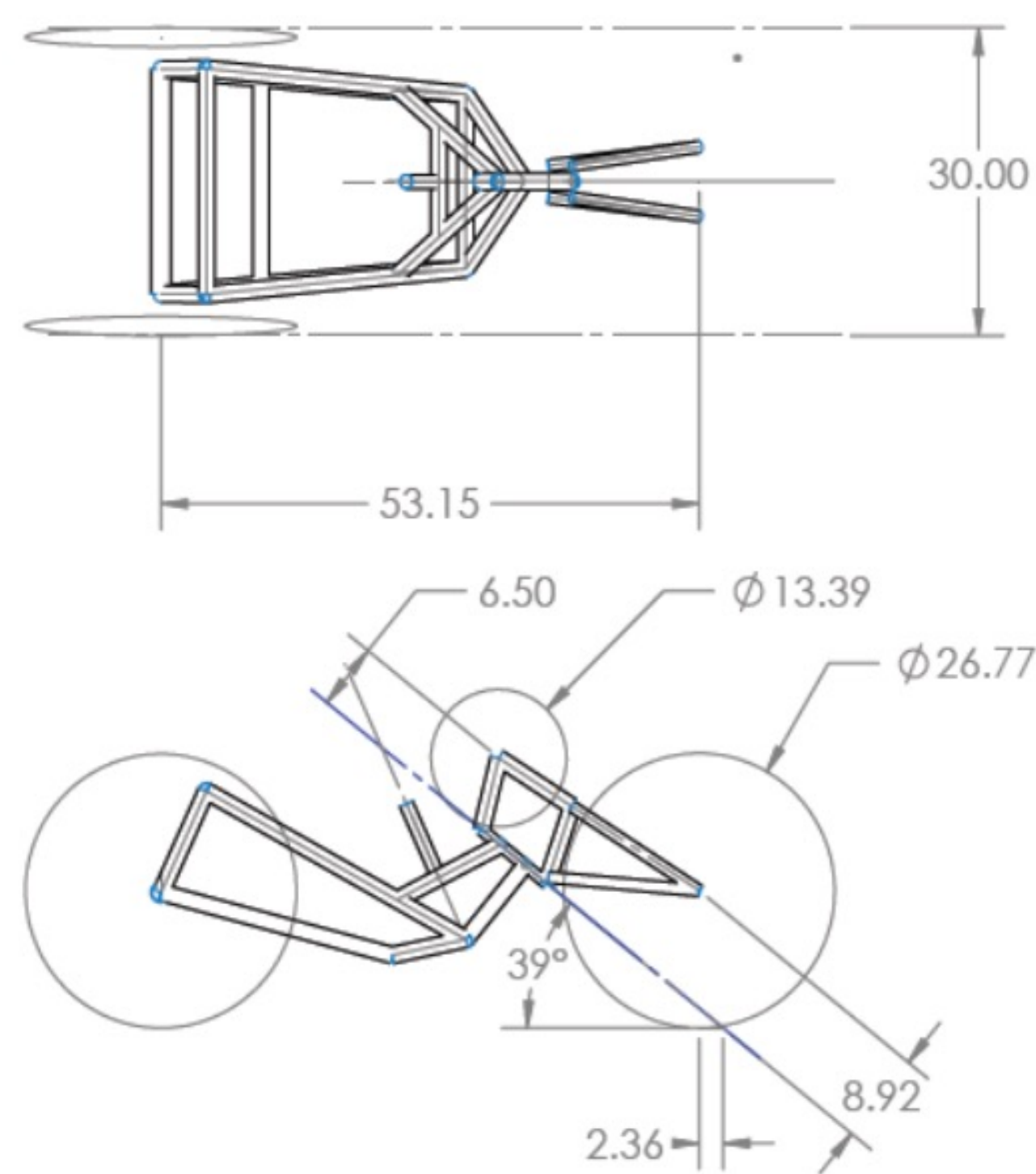
- SRAM GX Eagle Crankset, 1x12
- Cassette, Derailleur, Wireless Shifter
- SRAM DUB Bottom Bracket
- MRP Chain Guide
- Designed to mount with Universal Derailleur Hanger (UDH)
- PETG 3D printed chain guard with integrated spacing for chain guide



PARTNERSHIP PROGRAM COMPUTER SCIENCE & ENGINEERING

Partnership Program | Adaptive Sports Center | Jeni Blacklock | Greg Vanderbeek

MCEN4026: Manufacturing Processes and Systems 2022 - 2023



Goals

- Develop frame geometry
- Layout of all the parts

Lessons Learned

- Time consuming
- Limitations of software & skill
- Ergonomic data
- Group dynamics
- Project Management
- Communication

CAD Model

FDM Lugs &
PVC

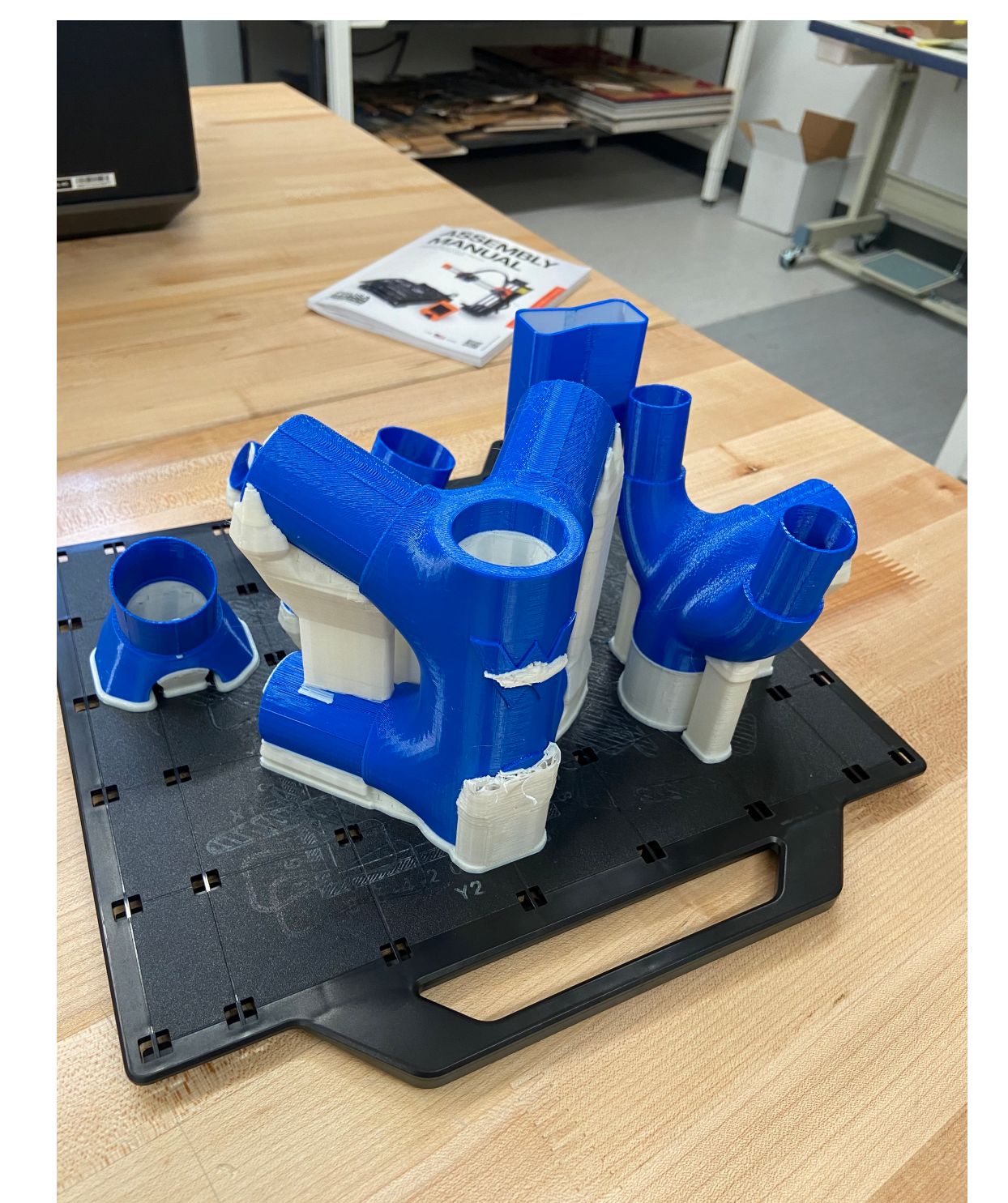
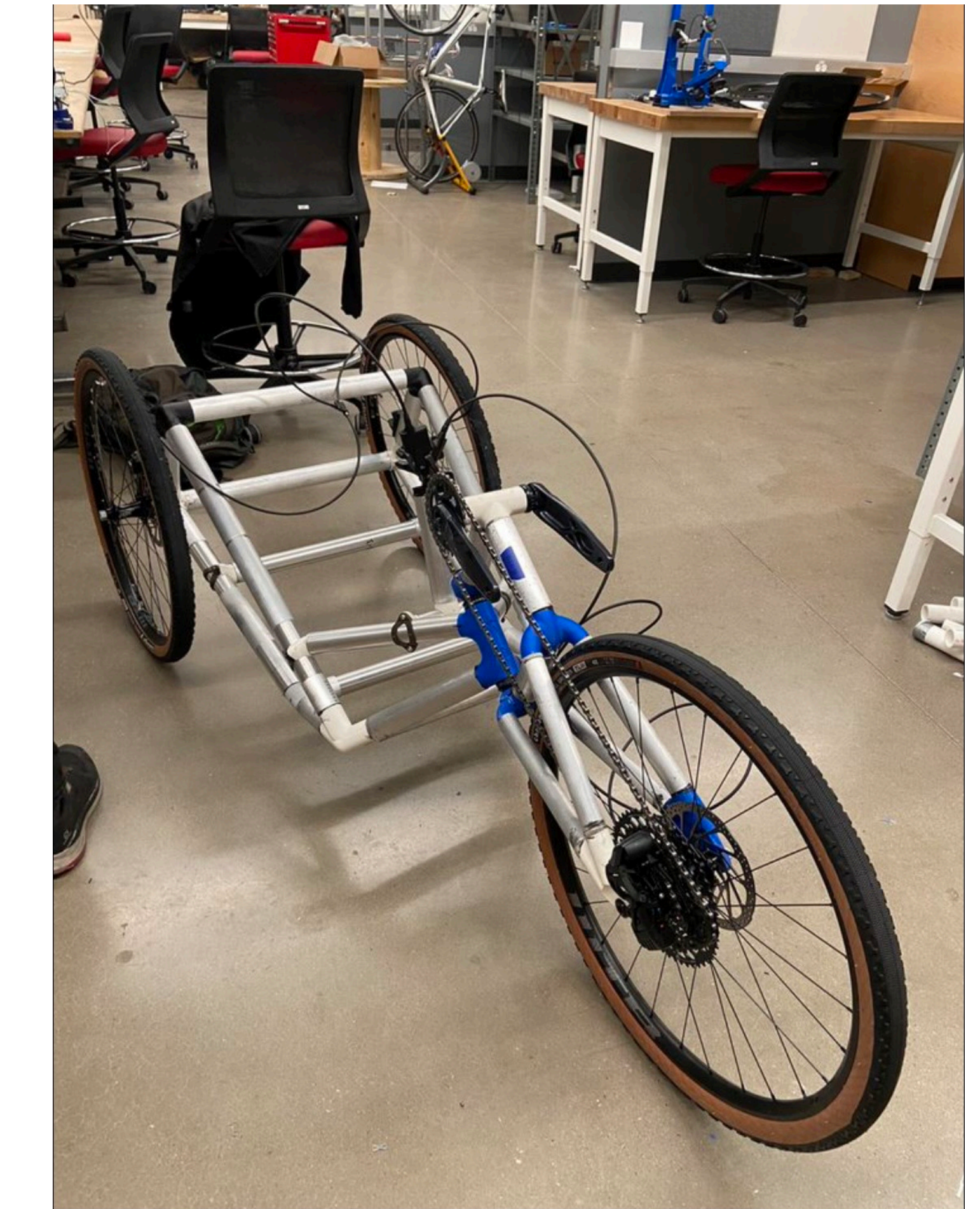
Titanium Lugs and
Carbon Fiber

Goals

- Verify frame sizing and geometry
- Check fitment of components
- Check alignment of drive train

Lessons Learned

- Models never line up exactly with the real world
- Rapid prototyping
- Manufacturing processes
- Project and deliverable timelines



Goals

- Manufacture final prototype

Lessons Learned

- Rushing to completion yields subpar results
- Taking time for the details is vital and alleviates obstacles near the finish

